Mid-Air Volumetric Display via Ultrasonically Induced Refractive-Index Gradients: Concept, Quantitative Feasibility, and Limitations of Opto-Acoustic Modulation in Free Air

Nitinkumar Vishwakarma

Department of Electronics and Telecommunication Engineering
Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology
Andheri, Mumbai, Maharashtra, India

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Abstract

Free-standing three-dimensional imagery that floats in open air, without screens, reflectors or head-mounted optics, remains one of the most compelling goals of display technology. Its popular image comes from science fiction, most famously the interactive holograms used by the character Tony Stark in Marvel films. Such imagery requires *addressable picture elements (voxels) in empty air*. This paper studies a concept in which those voxels are created acoustically. Interfering, phase-controlled ultrasonic waves from a transducer array create localised regions of compressed or rarefied air. Because the refractive index of a gas depends on its density, each region acts as a transient gradient-index optical element, and a synchronised laser beam fired through it is refracted (and diffracted) toward the viewer. The concept belongs to the established field of acousto-optics, which was recently extended to ambient air for high-power laser modulation.

We set out the physical principle and derive the chain from acoustic pressure to density, refractive index and optical deflection. We then quantify feasibility with present-day hardware. Phased ultrasonic arrays reach focal sound pressure levels of about 150–165 dB ($\approx$0.9–5 kPa peak), giving refractive-index perturbations of only $\delta n \approx 10^{-6}$–10^{-5}. The resulting ray deflections are $\sim$10–60 μrad, and first-order diffraction angles at 40 kHz are $\sim$60 μrad. Redirecting light by even 1° toward an off-axis viewer would require $\delta n \gtrsim 10^{-4}$, corresponding to $\gtrsim$190 dB. Larger angles exceed a hard physical ceiling: since density cannot be modulated by more than its own value, $|\delta n| \leq n_0 - 1 \approx 2.8 \times 10^{-4}$ in air.

We also present a control algorithm covering voxel scheduling, acoustic-hologram computation, phase-locked laser timing and viewer tracking, and estimate its computational budget, which fits within a single GPU/FPGA. The optical-deflection deficit of three to four orders of magnitude is identified as the decisive barrier. The limitations of the concept are stated explicitly, and applications and directions for experimental verification are outlined.

Keywords: volumetric display, mid-air display, acousto-optics, opto-acoustic modulation, ultrasonic phased array, refractive-index gradient, Gladstone–Dale relation, acoustic holography

1 Introduction

1.1 Motivation

Science fiction has shaped popular expectations of three-dimensional (3D) displays. The interactive, walk-around holograms used by Tony Stark in the Marvel films are the archetype: luminous, full-colour images suspended in open space that several people can view from any angle and manipulate with their bare hands. Despite the widespread use of the word "hologram", nearly all commercial products that claim this capability are 2D projections onto semi-transparent screens, beam-splitters (the "Pepper's ghost" illusion), rotating LED fans, or head-mounted displays. None of them places light at an arbitrary point in empty air.

1.2 The fundamental problem: creating a "pixel" in air

A true free-space volumetric display must create an addressable voxel at a chosen 3D location in a transparent medium, and that voxel must send light toward the observer's eye. Physically, only two families of mechanism can achieve this:

1. **Emission:** the medium at that point is made to emit light, e.g. by laser-induced plasma [1, 2].
2. **Redirection:** something at that point scatters, reflects or refracts light from an external source. Examples are a trapped particle [3, 4, 5] or, the subject of this paper, a *localised change in the optical properties of the air itself.*

The concept analysed here comes from a simple observation. Light changes direction when it crosses a region of different optical density (refraction), and sound waves *are* travelling regions of higher and lower density. If ultrasonic waves from several directions are made to interfere constructively at a chosen point, they produce a localised pocket of compressed (or rarefied) air. A laser beam passing through that pocket would be bent, and a correctly timed and aimed beam could be bent toward the viewer, making the pocket appear as a point of light.

Note on terminology. The bending of light at a change of optical density is *refraction*. *Dispersion* is the related dependence of refractive index on wavelength. The interaction of light with sound-induced index changes is known in the literature as *acousto-optic* interaction, and devices based on it are acousto-optic modulators and deflectors [6, 7, 8, 9, 10]. The term *opto-acoustic* (or *photoacoustic*) more commonly refers to the reverse effect, the generation of sound by absorbed light. This paper uses "acousto-optic" for the physics and "opto-acoustic modulation" as the project name.

The concept was conceived by the author independently, before learning of prior work. It turns out to be rooted in nearly a century of acousto-optics. Debye and Sears [6] and Lucas and Biquard [7] first observed diffraction of light by ultrasound in liquids in 1932, and Raman and Nath [8] explained it theoretically in 1935. Air was long considered too weak a medium for practical acousto-optics. However, Schrödel *et al.* [11] recently demonstrated efficient acousto-optic modulation of gigawatt-scale laser pulses in *ambient air*, using intense ultrasound in a resonant configuration. Meanwhile, ultrasonic phased arrays capable of steering focal points in 3D at kilohertz rates are now routine in mid-air haptics [12, 13] and acoustic levitation [14, 15, 16, 17].

1.3 Contributions

This paper:

- formalises the concept of an acousto-optic free-air volumetric display (Section 3);
- quantifies **how much density modulation is required** versus **what current ultrasonic sources can produce**, including fundamental (not merely technological) limits (Section 4);

- designs a **control and synchronisation algorithm** for voxel creation, laser firing and viewer-dependent rendering (Section 5);
- estimates the **computational requirements** (Section 6);
- surveys potential **applications** (Section 7);
- states the **limitations** of the concept explicitly (Section 8) and outlines future work and safety considerations (Sections 9 and 10).

2 Background and Related Work

2.1 Volumetric and mid-air displays

Volumetric displays have a long history [18, 19]. *Swept-volume* displays project images onto a rapidly rotating screen, while *static-volume* displays excite voxels inside a solid or gaseous medium. Free-air approaches include:

- **Laser-induced plasma displays.** Kimura *et al.* [2] focused nanosecond pulses to ionise air at scanned points, producing visible but loud and hot plasma voxels. Ochiai *et al.* [1] used femtosecond pulses and computer-generated holography to obtain safer, touchable voxels at up to ∼200,000 dots/s.
- **Photophoretic trap displays.** Smalley *et al.* [4] trapped a ∼10 μm cellulose particle in a photophoretic laser trap, scanned it rapidly, and illuminated it with RGB lasers to draw images by persistence of vision.
- **Acoustic trap displays.** Fushimi *et al.* [5] and Hirayama *et al.* [3] levitated a millimetre-scale bead with an ultrasonic phased array and scanned it at speeds of several metres per second. The Multimodal Acoustic Trap Display [3] simultaneously provided visual, tactile and audible content.

All of these displays rely on an emitter or a physical scatterer at the voxel. The concept studied here differs: the voxel is a *purely gaseous refractive-index structure*, with nothing introduced into the air.

2.2 Acousto-optics

A sound wave of pressure amplitude p produces a periodic modulation of the refractive index, which acts as a moving phase grating. Two diffraction regimes are distinguished by the Klein–Cook parameter $Q = 2\pi\lambda L/(n\Lambda^2)$, where λ is the optical wavelength, Λ the acoustic wavelength and L the interaction length [9, 10]. For $Q \ll 1$ the interaction is in the *Raman–Nath* regime, with many diffraction orders. For $Q \gg 1$ it is in the *Bragg* regime, with a single dominant order. Conventional acousto-optic devices use crystals or glasses driven at MHz–GHz frequencies, where δn is large and Λ is small. In air both factors are unfavourable. Nevertheless, Schrödel *et al.* [11] achieved >50 % diffraction efficiency using ∼500 kHz ultrasound at ∼145–150 dB in a standing-wave resonator, at the cost of small (sub-mrad) deflection angles.

2.3 Ultrasonic phased arrays

Arrays of tens to thousands of 40 kHz piezoelectric transducers, each individually phase-controlled, can synthesise focal points, twin traps and arbitrary multi-point fields [12, 13, 14, 15, 16, 17]. Single-focus phases follow directly from geometry, or equivalently from time reversal [20]. Multi-point fields require iterative phase retrieval, such as Gerchberg–Saxton-type algorithms [21]. GPU implementations such as GS-PAT compute ∼17,000 solutions per second for up to 32 focal points [17]. Typical focal sound pressure levels are ∼150–160 dB. Parametric "directional speakers" [22, 23] use the same transducers to beam audible sound, but at lower focal pressures.

3 Principle of Operation

3.1 System overview

The proposed system (Fig. 1) comprises:

1. one or more **ultrasonic phased arrays** surrounding or bounding the display volume;
2. an **FPGA-based driver** setting the phase and amplitude of each transducer;
3. a **laser source**, preferably pulsed or fast-modulated, in one or more visible wavelengths;
4. a **beam-steering unit** (galvanometer mirrors or acousto-optic deflectors);
5. a **viewer-tracking sensor** (camera or eye tracker);
6. a **host computer** running the rendering and synchronisation algorithm.

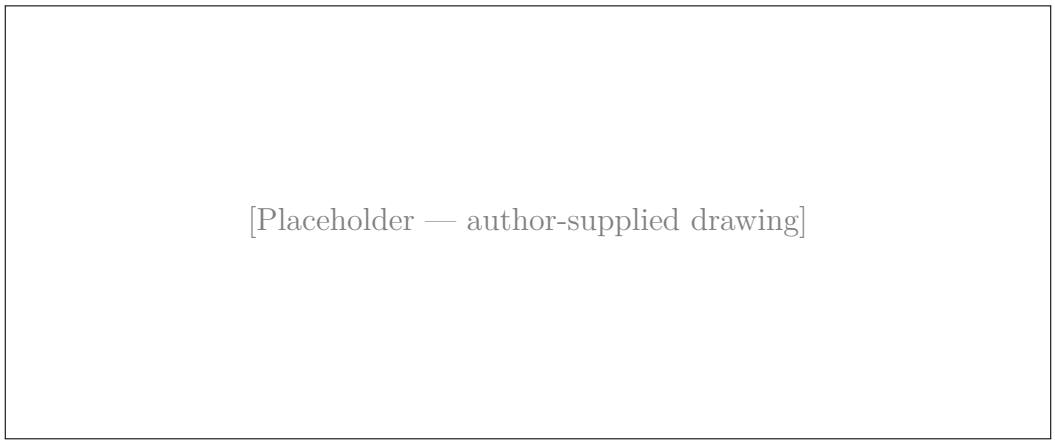


Figure 1: Diagram of the experimental setup: ultrasonic transducer array(s), focal region in the display volume, laser and beam-steering optics, observer position, and control electronics.

3.2 Creating a localised density perturbation by interference

Two counter-propagating plane waves of equal amplitude p_a and wavenumber $k = 2\pi/\Lambda$ superpose into a standing wave:

$$p(x,t) = p_a \cos(kx - \omega t) + p_a \cos(kx + \omega t) = 2p_a \cos(kx)\cos(\omega t). \tag{1}$$

This produces stationary pressure antinodes, where the density oscillates with twice the single-wave amplitude, separated by $\Lambda/2$ (Fig. 2). With an array of N transducers at positions $\mathbf{r}_i$, each element is driven with phase

$$\phi_i = -k\,|\mathbf{r}_f - \mathbf{r}_i| \pmod{2\pi}, \tag{2}$$

so that all contributions arrive in phase at the chosen focus $\mathbf{r}_f$. The focal pressure then grows approximately linearly with the number of elements that "see" the focus. The focal spot has a lateral width of about $\Lambda/2$ ($\approx$4.3 mm at 40 kHz) and is elongated along the array axis.

3.3 From pressure to density to refractive index

For small-amplitude, adiabatic sound, the density perturbation is [24]

$$\delta\rho = \frac{p}{c^2}, \tag{3}$$

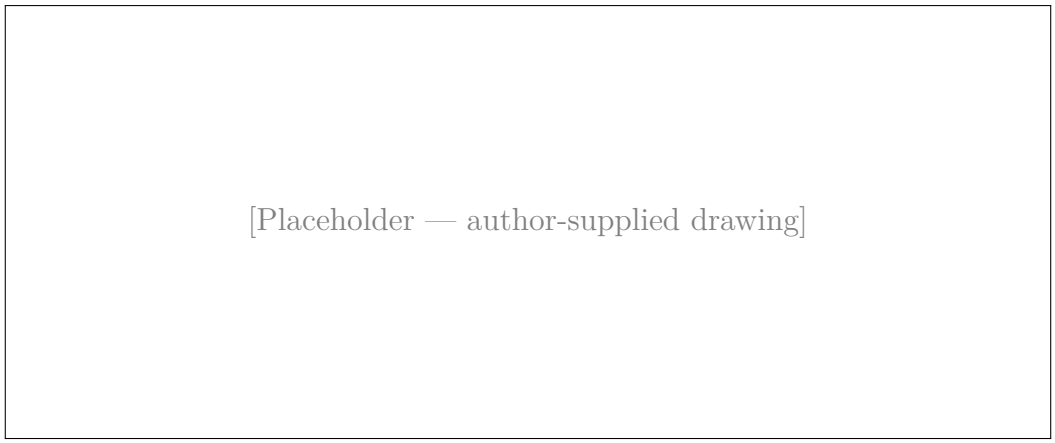


Figure 2: Two ultrasonic waves interfering to create regions of high density (compression) and low density (rarefaction).

with $c \approx 343\,\mathrm{m/s}$ in air at $20\,^{\circ}\mathrm{C}$. The refractive index of a gas follows the Gladstone–Dale relation [25]:

$$n - 1 = K\rho, \qquad K \approx 2.31 \times 10^{-4}\ \mathrm{m^3/kg} \quad (\text{air, } 532\ \mathrm{nm}), \tag{4}$$

using $n_0 - 1 \approx 2.78 \times 10^{-4}$ [26] and $\rho_0 = 1.204\,\mathrm{kg/m^3}$. Combining (3) and (4) gives the *piezo-optic coefficient* of air:

$$\frac{\delta n}{p} = \frac{K}{c^2} = \frac{n_0 - 1}{\gamma p_0} \approx 1.96 \times 10^{-9}\ \mathrm{Pa^{-1}}. \tag{5}$$

Dispersion makes K vary by only $\sim$1–2 % across the visible spectrum [26], so the effect is nearly the same for red, green and blue lasers.

3.4 Deflection of light: ray picture

In a medium with a refractive-index gradient, a ray obeys $\frac{d}{ds}\left(n\,\frac{d\mathbf{r}}{ds}\right) = \nabla n$ [27]. For small deflections, the angular deviation after an interaction length L is

$$\theta \approx \int_L \nabla_\perp n \, ds \approx L\,|\nabla_\perp n|. \tag{6}$$

In a sinusoidal field the peak gradient is $|\nabla n|_{\max} = 2\pi\,\delta n/\Lambda$. With $L \approx \Lambda$ (the focal length scale), the deflection becomes

$$\theta \approx 2\pi\,\delta n. \tag{7}$$

This result is remarkably simple: **the deflection angle in radians is of the order of the index perturbation itself, multiplied by** 2π.

3.5 Deflection of light: wave picture

The same field imposes an optical phase modulation of amplitude

$$\varphi = \frac{2\pi}{\lambda}\,\delta n\, L. \tag{8}$$

At 40 kHz, $Q = 2\pi\lambda/\Lambda \approx 4 \times 10^{-4} \ll 1$ (with $L \approx \Lambda$), so the interaction lies deep in the Raman–Nath regime. The diffraction orders appear at $\theta_m = m\lambda/\Lambda$ with efficiency $\eta_m = J_m^2(\varphi)$, where J_m is the Bessel function of order m [8, 9, 28].

3.6 Synchronisation with the acoustic cycle

The density pattern oscillates at the acoustic frequency (period 25 μs at 40 kHz). A *pulsed* or *gated* laser fired at a fixed acoustic phase therefore "freezes" the structure (stroboscopic operation). A CW laser, by contrast, averages over the cycle, and its mean deflection cancels. The timing of laser firing is therefore a core part of the control algorithm (Section 5).

4 Quantitative Feasibility Analysis

All numbers below are computed by the accompanying script `feasibility_calc.py` (Appendix A) for air at 20 °C, 1 atm, $\lambda = 532$ nm and $f = 40$ kHz unless stated otherwise. SPL is quoted re 20 μPa (rms). The tables give peak pressure amplitude.

4.1 What current ultrasonic sources produce

Table 1: Representative focal pressures of airborne ultrasonic sources.

Source	Typical focal SPL	Peak pressure
Single 40 kHz transducer (10 mm, 10 V_{pp}) at 30 cm	∼120 dB	∼28 Pa
Parametric directional speaker (on-axis, near field)	∼120–135 dB	∼30–160 Pa
Mid-air haptic array (∼256 elements) [12, 13]	∼150–155 dB	∼0.9–1.6 kPa
Levitation / standing-wave arrays [14, 15, 16]	∼155–165 dB	∼1.6–5 kPa
Air acousto-optic modulator (500 kHz, resonant) [11]	∼145–150 dB	∼0.5–0.9 kPa

4.2 Resulting refractive-index perturbation and deflection

Table 2: Index perturbation, ray deflection (Eq. 7) and optical phase modulation (Eq. 8) versus focal SPL.

SPL (dB)	p_{peak} (Pa)	$\delta\rho/\rho_0$	δn	θ (μrad)	φ (rad)
120	28	2.0×10^{-4}	5.6×10^{-8}	0.35	0.006
140	283	2.0×10^{-3}	5.6×10^{-7}	3.5	0.056
150	894	6.3×10^{-3}	1.8×10^{-6}	11	0.18
155	1,591	1.1×10^{-2}	3.1×10^{-6}	20	0.32
160	**2,828**	$\mathbf{2.0 \times 10^{-2}}$	$\mathbf{5.6 \times 10^{-6}}$	**35**	**0.56**
165	5,030	3.6×10^{-2}	9.9×10^{-6}	62	1.00
170	8,944	6.3×10^{-2}	1.8×10^{-5}	110	1.78

At a representative state-of-the-art focal level of 160 dB, the air density changes by about 2 %, and a ray crossing the focus is bent by about **35 μrad (0.002°)**. At a viewing distance of 0.5 m, this shifts the beam by ∼17 μm. The angle is about eight times *smaller* than the angular resolution of the human eye (∼1 arcmin ≈ 290 μrad).

The wave picture is somewhat more encouraging in one respect. At 160 dB, $\varphi \approx 0.56$ rad, so roughly $J_1^2(0.56) \approx 7\,\%$ of the light goes into each first diffraction order. A meaningful *fraction* of the beam is therefore redirected, but only by $\theta_1 = \lambda/\Lambda \approx 62\,\mu$rad. Both pictures agree: **the light is affected, but it barely changes direction.**

4.3 How much density modulation is required?

For the voxel to be visible, light must reach an observer who is generally *not* looking down the laser beam. If the observer sits at angle α from the beam axis, the voxel must deflect light by α.

A useful display needs α of at least several degrees, and ideally tens of degrees for multi-viewer, walk-around viewing.

(a) Optimistic upper bound: grazing incidence on a sharp index step. The largest deviation a single index step Δn can produce occurs at grazing incidence (total internal reflection), $\theta_{\max} \approx \sqrt{2\Delta n}$ (Table 3).

Table 3: Index contrast required for a given deflection (grazing-incidence upper bound).

Desired deflection	Required Δn	Required p	Equivalent SPL	$\Delta n/(n_0 - 1)$
0.01°	1.5×10^{-8}	7.8 Pa	109 dB	0.0001
0.1°	1.5×10^{-6}	776 Pa	149 dB	0.005
0.5°	3.8×10^{-5}	19 kPa	177 dB	0.14
1°	1.5×10^{-4}	78 kPa	189 dB	0.55
5°	3.8×10^{-3}	1.9 MPa	217 dB	**13.7 (impossible)**
10°	1.5×10^{-2}	7.8 MPa	229 dB	**55 (impossible)**

Even this generous bound is unrealistic. Grazing incidence on a ∼2 mm wide gradient at 0.1° requires a path of ∼1 m along the interface, far longer than an acoustic focus.

(b) Realistic estimate: gradient-index (GRIN) deflection over a compact focal region. Using Eq. (6) with $L = 1$ cm and the index change occurring over $\Lambda/4 \approx 2.1$ mm gives Table 4.

Table 4: Required index gradient for GRIN deflection ($L = 1$ cm).

Desired deflection	Required $\lvert\nabla n\rvert$	Required Δn over $\Lambda/4$	Relative to $(n_0 - 1)$ of air
0.1°	$0.17\,\mathrm{m}^{-1}$	3.7×10^{-4}	1.3×
1°	$1.7\,\mathrm{m}^{-1}$	3.7×10^{-3}	13×
10°	$17\,\mathrm{m}^{-1}$	3.7×10^{-2}	135×

4.4 A fundamental ceiling

Density cannot fall below zero, and compressions beyond $\delta\rho \sim \rho_0$ involve strongly nonlinear, shock-dominated acoustics. Hence, *independent of transducer technology*,

$$|\delta n| \lesssim n_0 - 1 \approx 2.8 \times 10^{-4} \qquad \text{(air, STP)}. \tag{9}$$

Inserting this into Eq. (6) with $L = 1$ cm and a $\Lambda/4$ gradient gives a ceiling of $\theta_{\max} \approx 1.3$ mrad (0.07°). An extended 10 cm interaction region raises it to only ≈0.7°. **Pure refraction in ambient air cannot redirect a laser beam by more than about one degree, no matter how powerful the sound source.**

4.5 Can a higher acoustic frequency help?

In the diffraction picture, the deflection angle λ/Λ does not depend on δn, so one might raise the ultrasonic frequency to shrink Λ (Table 5).

Atmospheric sound absorption, however, rises roughly with the square of frequency [29]. It is of order 1 dB/m at 40 kHz, tens of dB/m at several hundred kHz, and far higher at MHz frequencies, so a 100 MHz sound field cannot propagate meaningfully through air. Higher frequencies also make high SPL harder to generate and lower the shock-formation threshold.

Table 5: First-order diffraction angle versus acoustic frequency ($\lambda = 532\,\text{nm}$).

Frequency	Λ	First-order angle
40 kHz	8.6 mm	0.062 mrad (0.004°)
100 kHz	3.4 mm	0.16 mrad (0.009°)
500 kHz	0.69 mm	0.78 mrad (0.044°)
2 MHz	0.17 mm	3.1 mrad (0.18°)
113 MHz	**3 μm**	**10°**

4.6 Nonlinear and secondary effects

- **Shock formation.** The plane-wave shock-formation distance $\bar{x} = \rho_0 c^3/(\beta\omega p)$ [30] at 40 kHz is ∼18 cm at 150 dB, ∼5.7 cm at 160 dB and ∼1.8 cm at 170 dB. Above ∼160 dB, energy cascades into harmonics, saturating the achievable focal pressure (acoustic saturation).
- **Acoustic streaming and radiation pressure.** Intense focal fields drive steady air flows and exert forces on dust and droplets [31, 32], disturbing the purely gaseous voxel.
- **Adiabatic heating.** At 160 dB the peak temperature excursion is $\Delta T = T_0 \frac{\gamma-1}{\gamma}\frac{p}{p_0} \approx 2.3\,\text{K}$. It oscillates at 40 kHz and adds an index change that is negligible compared with Eq. (5).
- **Ambient turbulence and temperature drift.** A 1 °C change alters c by ∼0.6 m/s, shifting the acoustic phase across a 20 cm path by ≈0.26 rad at 40 kHz. Natural convection produces $\delta n \sim 10^{-6}$ fluctuations, comparable to the signal itself (the basis of schlieren imaging [33]).

4.7 Can density modulation instead modulate scattering?

An alternative is to use the voxel to change how much light the air *scatters* rather than to steer the beam. Clean air scatters light by Rayleigh scattering (extinction $\approx 1.3 \times 10^{-5}\,\text{m}^{-1}$ at 532 nm [34]), proportional to molecular number density. Consider a 1 W, 532 nm beam crossing a 4.3 mm voxel at 160 dB (2 % density modulation), viewed from 0.5 m with a 5 mm pupil:

- total power scattered by the voxel: $5.6 \times 10^{-8}\,\text{W}$;
- power entering the eye: $3.5 \times 10^{-13}\,\text{W}$ ($\sim 9 \times 10^5$ photons/s);
- *modulated* (signal) part: $7 \times 10^{-15}\,\text{W}$ ($\sim 2 \times 10^4$ photons/s).

The signal is detectable by a dark-adapted eye. However, the *entire* 1 m beam also scatters, giving a background ∼12,000× larger than the voxel's signal, i.e. a contrast of ∼0.01 %. The viewer would see a faint laser line, not a voxel.

4.8 Summary of the feasibility gap

Table 6: Achievable versus required quantities.

Quantity	Achievable today (160 dB)	Required ($\geq 10°$ viewing)	Gap
δn	5.6×10^{-6}	$\gtrsim 10^{-2}$	∼2,000–5,000×
Deflection	35 μrad	175 mrad	∼5,000×
Physical ceiling	—	exceeds $(n_0 - 1)$ by 50–135×	**fundamental**

Present directional ultrasonic sources produce density gradients about three to four orders of magnitude too weak to redirect light to off-axis observers. Moreover, ambient air cannot support the required index contrast even in principle, so technical improvements alone (larger arrays, more power) cannot close the gap. The analysis does confirm that the acoustically induced

optical effect is real and measurable: phase modulations of order 1 rad are achievable, and this is precisely what has been exploited for laser beam modulation in air [11].

5 Control and Synchronisation Algorithm

An opto-acoustic display needs a real-time pipeline that decides *which* voxels to create, *where* and *when* to create them, and *when and in which direction* to fire the laser.

5.1 Pipeline overview

```
3D content (mesh / CAD / point cloud / volumetric data)
       |
[1] Voxelisation & sampling  --> voxel list {position, colour, intensity, priority}
[2] Viewer tracking          --> eye positions e_j (view-dependent rendering)
[3] Temporal scheduling      --> ordered sequence of acoustic frames (M foci each)
[4] Acoustic hologram solver --> transducer phases/amplitudes per frame
[5] Optical path planner     --> laser direction, wavelength, power per voxel
[6] Phase-locked timing      --> FPGA: update array, fire laser at t_fire
[7] Closed-loop calibration  <-- microphones / schlieren camera / temperature
```

5.2 Stage 1 — Voxelisation

The 3D model is converted into a set of voxel targets. For surface models, Poisson-disk sampling of the mesh surface gives near-uniform spacing. The minimum useful spacing is set by the focal spot size ($\sim \Lambda/2 \approx 4.3\,\text{mm}$ at 40 kHz). Back-face culling relative to the tracked viewer removes hidden voxels. Each voxel carries a priority (silhouette edges and interaction points first), which enables graceful degradation when the voxel budget is exceeded.

5.3 Stage 2 — Viewer tracking

A scattering voxel, such as a particle or plasma, radiates in all directions. A *refractive* voxel only redirects the beam into a narrow cone whose width is set by the diffraction of the beam itself. Each voxel must therefore be rendered *per eye*: given eye position $\mathbf{e}$, voxel position $\mathbf{v}$ and deflection capability $\theta(\mathbf{v})$, the laser must enter along a direction within θ of $(\mathbf{e}-\mathbf{v})$. The display thus behaves like a light-field display. Stereo viewing requires rendering each voxel twice (once per eye), and the number of laser passes scales with the number of viewers.

5.4 Stage 3 — Temporal scheduling

Table 7: Relevant timescales.

Timescale	Value
Acoustic period (40 kHz)	25 μs
Acoustic time of flight across 20 cm	0.58 ms
Transducer ring-up / array update interval	$\sim$0.1–1 ms (1–10 kHz [17, 3])
Persistence of vision (flicker-free refresh)	$\leq$33 ms ($\geq$30 Hz)

With array update rate R_u, M simultaneous foci per frame and refresh rate R_f, the voxel budget per refresh is

$$N_{\text{vox}} = \frac{M\,R_u}{R_f}. \tag{10}$$

For example, $R_u = 10\,\text{kHz}$, $M = 32$ and $R_f = 30\,\text{Hz}$ give $\sim$10,000 voxels per refresh. However, splitting the array's output among M foci reduces the pressure at each focus to roughly $1/M$,

and hence $\delta n \propto 1/M$. This creates a hard *brightness–resolution trade-off* that compounds the deficit identified in Section 4. Scheduling is therefore formulated as an optimisation. Voxels are ordered to minimise focus displacement between frames (a travelling-salesman-like problem, solved approximately with nearest-neighbour plus 2-opt heuristics), and spatially separated voxels are grouped into the same frame to reduce cross-talk between foci.

5.5 Stage 4 — Acoustic hologram solver

- **Single focus:** phases are given in closed form by Eq. (2), at $O(N)$ cost.
- **Multiple foci:** an iterative Gerchberg–Saxton / weighted-GS scheme [17, 21] alternates between the transducer plane, where the amplitude constraint $|u_i| \le u_{\max}$ is enforced, and the focal points, where the desired amplitudes are enforced with free phase. The propagation matrix $\mathbf{F} \in \mathbb{C}^{M\times N}$, built from a piston-source directivity model, is precomputed. Each iteration costs two matrix–vector products.

5.6 Stage 5 — Optical path planner

For each voxel and each eye, the planner computes a laser entry direction that (i) passes through the voxel, (ii) lies within the deflection cone toward the eye, (iii) avoids passing through other active foci, preventing unintended deflections, and (iv) respects laser-safety zones (Section 10). It also selects wavelength and power for colour and brightness. Galvanometer scanners ($\sim$10–30 k points/s) are adequate for small voxel counts; acousto-optic deflectors (MHz rate) or spatial light modulators [1] are needed for larger scenes.

5.7 Stage 6 — Phase-locked laser timing

The laser must be fired when the acoustic field at the voxel has the desired phase (maximum gradient or maximum compression):

$$t_{\text{fire}} = t_{\text{update}} + \max_i \frac{|\mathbf{r}_f - \mathbf{r}_i|}{c} + t_{\text{settle}} + \frac{\psi_{\text{target}} - \psi_f}{\omega} + kT, \quad k \in \mathbb{N}, \tag{11}$$

where ψ_f is the field phase at the focus and T the acoustic period. A timing error δt reduces the effective modulation by a factor $\cos(\omega\,\delta t)$. Keeping losses below 5 % requires $\delta t \lesssim 1.3\,\mu$s at 40 kHz, which is easily achievable with an FPGA sharing a common clock between the transducer drivers and the laser trigger. The pulse length should be much shorter than 25 μs; nanosecond-to-microsecond pulses are suitable.

5.8 Stage 7 — Closed-loop calibration

Because $\delta n \sim 10^{-6}$ is comparable to thermal turbulence, open-loop operation is unreliable. A calibration loop should include:

- a temperature/humidity sensor updating $c(T, RH)$ in Eq. (2) ($\approx$0.26 rad phase error per °C over 20 cm);
- periodic focal-field mapping with a scanned microphone or a schlieren/shadowgraph camera [33];
- per-transducer phase/amplitude correction to compensate manufacturing spread (typically $\pm$10–20°).

5.9 Pseudocode

```
loop every refresh period (<= 33 ms):
    eyes   <- tracker.get_eye_positions()
    voxels <- voxelise(scene, eyes)                  # cull, prioritise
    frames <- schedule(voxels, M, R_u, R_f)          # group into M-focus frames
    for frame in frames:
        phases <- solve_hologram(frame.foci, F, iters)    # GS-PAT-style, GPU
        fpga.queue_array_update(phases, t_update)
        for v in frame.foci:
            for e in eyes:
                dir, lam, P <- plan_laser(v, e, active_foci=frame.foci)
                t <- compute_fire_time(v, phases, t_update)   # Eq. (11)
                fpga.queue_laser_pulse(dir, lam, P, t)
    calibrate_if_due()
```

6 Computational Requirements

Assuming ∼8 floating-point operations per complex multiply–accumulate and one forward plus one backward propagation per iteration, the hologram solver costs are given in Table 8.

Table 8: Acoustic-hologram solver cost.

Array size N	Foci M	Iterations	FLOP/solution	At 10,000 solutions/s
256	1	closed form	4×10^3	0.04 GFLOP/s
256	32	10	1.3×10^6	13 GFLOP/s
1,024	32	10	5.2×10^6	52 GFLOP/s
4,096	100	20	1.3×10^8	1.3 TFLOP/s

Other tasks are lighter:

- **Voxelisation and scheduling:** 10^4–10^5 voxels per frame at 30 Hz with $O(n \log n)$ heuristics, i.e. a few GFLOP/s.
- **Viewer tracking:** off-the-shelf eye/face tracking at 60–120 Hz, requiring a few GFLOP/s or a dedicated accelerator.
- **Optical path planning:** $O(N_{\text{vox}} \times N_{\text{eyes}})$ geometric queries, negligible compared with the solver.
- **Hard real-time control:** the FPGA must handle $N \times R_u \times$(phase + amplitude bits), e.g. $4096 \times 10\,\text{kHz} \times 16\,\text{bit} \approx 650\,\text{Mbit/s}$, and generate μs-precision triggers.

Even the largest configuration (∼1.3 TFLOP/s) is within reach of a single modern consumer GPU, which delivers tens of TFLOP/s, paired with a mid-range FPGA. Computational power is a design consideration but **not** a limiting factor for this technology. The binding constraint is the physics quantified in Section 4.

7 Applications

If a mature free-air volumetric display could be realised, potential applications would include:

1. **Engineering and CAD.** Designers and engineering teams could review 3D models at true scale and depth from any angle without headsets. Combined with hand tracking, components could be rotated, exploded, sectioned and assembled in mid-air. Ultrasonic phased arrays already provide *mid-air haptic feedback* [12, 13, 35], so the same hardware could let users "feel" surfaces and constraints.

2. **3D scientific and data visualisation.** Molecular structures, fluid-flow fields, geological strata and multi-dimensional datasets could be explored with genuine depth cues (accommodation and convergence agree), avoiding the visual fatigue of stereoscopic screens.
3. **Medical imaging and surgical planning.** CT/MRI reconstructions could be shown as floating anatomy for a surgical team to inspect together. A touchless display is also attractive in sterile environments.
4. **Education and museums.** Floating models of planets, organs, machines or archaeological artefacts would allow many students to observe simultaneously from different positions.
5. **Entertainment and live events.** Concerts, theatre, theme parks and gaming could use effects that appear to occupy real space, extending current laser-show and projection-mapping techniques.
6. **Telepresence and live meetings.** Volumetric capture of a remote participant could be reproduced as a life-like 3D presence, supporting natural eye contact and gesture.
7. **Automotive and aviation interfaces.** Mid-air dashboards and head-up indicators would need no physical screens, with tactile confirmation from ultrasonic haptics.
8. **Command-and-control and situational awareness.** 3D "sand tables" for air-traffic control, disaster response or defence planning would let teams share a common operational picture.
9. **Retail, advertising and public information.** Eye-catching floating product displays and wayfinding signage could replace physical screens in public spaces.
10. **Hygienic and accessible public interfaces.** Touchless kiosks, lift controls and ATMs would reduce surface contact, with combined visual, tactile and audio feedback [3] aiding users with visual impairment.

Most of these applications assume wide viewing angles and multiple simultaneous viewers. As Section 4 shows, these are exactly the conditions that pure refractive deflection in air cannot currently meet (Section 8).

8 Limitations

The concept presented in this paper is incomplete, and it cannot yet be realised as a practical display. Its principal limitations are summarised below.

L1. Insufficient density gradient from current sources. State-of-the-art phased arrays (150–165 dB) produce $\delta n \approx 10^{-6}$–10^{-5} and deflections of 10–60 μrad. Off-axis viewing at $\geq 10^\circ$ requires roughly 10^3–10^4 times more (Table 6).

L2. Fundamental refractive-index ceiling of air. Even with unlimited acoustic power, $|\delta n| \lesssim n_0 - 1 \approx 2.8 \times 10^{-4}$ (Eq. 9). This limits refractive deflection in ambient air to $\sim$1 mrad for a compact focus and $< 1^\circ$ for any realistic interaction length. The limitation is physical, not technological.

L3. Frequency–attenuation trade-off. Larger diffraction angles require shorter acoustic wavelengths, but atmospheric absorption grows roughly with the square of frequency. A 10° diffraction angle would need $\sim$113 MHz sound, which cannot propagate in air.

L4. Acoustic nonlinearity and saturation. Above $\sim$160 dB, shock formation within a few centimetres transfers energy to harmonics and caps the achievable focal pressure.

L5. Narrow, view-dependent emission. A refractive voxel redirects light into a narrow cone rather than radiating in all directions. Each voxel must be rendered per eye and per viewer

using viewer tracking, which restricts the number of simultaneous observers and complicates stereo viewing.

L6. Brightness–resolution trade-off. Distributing the array output among M simultaneous foci reduces δn at each focus by $\sim 1/M$, worsening L1 as voxel count increases.

L7. Poor contrast of scattering-based readout. Using density modulation to modulate Rayleigh scattering yields a voxel-to-background contrast of only $\sim$0.01 % (Section 4).

L8. Environmental sensitivity. Natural convection produces index fluctuations ($\sim 10^{-6}$) comparable to the acoustic signal, and temperature drift shifts the acoustic phase by $\sim$0.26 rad/°C over 20 cm, so continuous calibration is required.

L9. Safety constraints. A refraction-based voxel by design directs laser light toward the viewer's eye, and the focal ultrasound levels far exceed occupational exposure guidelines (Section 10).

L10. Absence of experimental validation. All results in this paper are analytical, order-of-magnitude estimates based on linear acoustics, the Gladstone–Dale relation and idealised focal geometries. No prototype has yet been built or measured. Real focal fields, nonlinear effects and beam profiles may change the numbers, although they are not expected to change the conclusions.

9 Future Work

The following steps would test and refine the analysis of the concept as proposed:

1. **Measure δn at an acoustic focus.** A 16×16, 40 kHz array and a Mach–Zehnder interferometer or schlieren system [33] can verify Eq. (5) and Table 2 (expected $\varphi \approx 0.2$–0.6 rad).
2. **Phase-locked strobe imaging.** A pulsed laser synchronised by Eq. (11) can demonstrate a stationary, phase-resolved image of the focal index structure.
3. **Deflection versus SPL characterisation.** The beam deflection and diffraction efficiency should be measured as functions of focal SPL, frequency and interaction length, and compared with Tables 2–5.
4. **Hardware scaling studies.** The achievable focal pressure can be quantified with larger and denser arrays, resonant enhancement using acoustic reflectors [11], higher-power and wider-bandwidth transducers, and enclosed gases with a larger piezo-optic coefficient (e.g. sulphur hexafluoride, roughly 3–4× that of air).
5. **Algorithm implementation.** The pipeline of Section 5 can be implemented in simulation and on FPGA/GPU hardware to validate scheduling, hologram solving and μs-level laser timing.

10 Safety Considerations

- **Laser safety.** A refraction-based voxel works by directing a laser beam *toward the viewer's eye*. Light reaching the pupil must remain within the maximum permissible exposure defined by IEC 60825-1 [36], effectively Class 1/2 levels ($\lesssim$1 mW visible for CW) at the eye position. The optical path planner (Section 5) must enforce exclusion zones and power limits.
- **Airborne ultrasound exposure.** Focal levels of 150–165 dB far exceed commonly cited occupational guidelines for airborne ultrasound (roughly 110–115 dB in the 20–50 kHz

bands [37]). Even though the focus is localised and inaudible, the display volume should not overlap with users' heads. Hands may enter, as is routine in mid-air haptics.

11 Discussion

The concept examined here has a sound physical basis. Ultrasonic interference *does* create localised density pockets, and these *do* modify the propagation of light, as acousto-optics has established since 1932 [6, 7, 8] and as was recently demonstrated in ambient air at gigawatt optical powers [11]. The quantitative analysis, however, separates two very different questions:

1. *Can sound in air measurably change a laser beam?* **Yes.** Phase modulations approaching 1 rad and diffraction efficiencies of several percent (tens of percent with resonant enhancement [11]) are achievable.
2. *Can sound in air redirect a laser beam by angles large enough to reach a viewer standing to the side?* **Not in ambient air with present understanding.** Achievable deflections are $\sim 10^{-5}$ rad, and the physical ceiling of Eq. (9) is $\sim 10^{-3}$ rad.

This distinction explains why air acousto-optics has so far been developed as a *modulator* for high-power laser beams rather than as a display. For the remaining two challenges, synchronisation and computation, the analysis is encouraging. The algorithmic pipeline builds on established techniques (acoustic holography, phase-locked triggering, scheduling heuristics, eye tracking), and the computational load fits comfortably on commodity hardware.

12 Conclusion

We have formalised and quantitatively evaluated a concept for a free-air volumetric display based on opto-acoustic (acousto-optic) modulation. Interfering ultrasonic waves create localised refractive-index gradients that deflect a synchronised laser beam toward the observer.

Using the Gladstone–Dale relation and linear acoustics, we showed that the 150–165 dB focal fields of current phased arrays produce index perturbations of 10^{-6}–10^{-5} and deflections of 10–60 μrad, roughly 10^3–10^4 times less than practical off-axis viewing requires. A fundamental ceiling, $|\delta n| \lesssim n_0 - 1$, limits refractive deflection in air to about a milliradian regardless of acoustic power.

We designed a complete control pipeline covering voxelisation, viewer tracking, scheduling, multi-focus hologram solving, phase-locked laser timing and closed-loop calibration. Its computational cost is at most $\sim$1 TFLOP/s, within the capability of a single GPU and FPGA.

The decisive challenge is therefore physical rather than computational. The concept is incomplete, and its limitations have been stated explicitly. We hope this analysis provides a clear, quantitative foundation for future experimental work on opto-acoustic modulation in free air.

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[To be added by the author.]

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A Reproducing the Calculations

All tabulated values in Sections 4 and 6 can be reproduced by running `python feasibility_calc.py`. Key assumptions: air at $20\,^{\circ}\mathrm{C}$ and $1\,\mathrm{atm}$; $n_0 - 1 = 2.78 \times 10^{-4}$ at $532\,\mathrm{nm}$; $c = 343\,\mathrm{m/s}$; $\gamma = 1.40$; SPL re $20\,\mu\mathrm{Pa}$ rms; peak pressure $= \sqrt{2}\,p_{\mathrm{rms}}$; interaction length $L \approx \Lambda$ for Table 2 and $L = 1\,\mathrm{cm}$ for the GRIN estimate; Rayleigh extinction $1.3 \times 10^{-5}\,\mathrm{m}^{-1}$.